S.Y. B. Tech. Academic Year 2026-27 Semester: III

CSE20350/AID20210/CLD 20020: Data Structures Laboratory

WRITE UP

Experiment Number: 01

TITLE: Matrix Operations in C

PROBLEM STATEMEN:
To write a C program to perform basic matrix operations such as addition, subtraction, multiplication, and transpose of a matrix using two-dimensional arrays.

Objectives

- To understand the concept of two-dimensional arrays in C.
- To perform arithmetic operations on matrices.
- To implement matrix addition and subtraction.
- To implement matrix multiplication using nested loops.
- To find the transpose of a matrix.
- To develop programming skills using arrays and loops.

Theory
A matrix is a rectangular arrangement of elements organized in rows and columns. In C programming, matrices are represented using two-dimensional arrays. Matrix operations are widely used in computer graphics, image processing, machine learning, scientific computing, and engineering applications.

1. Matrix Addition
Condition: Both matrices must have the same number of rows and columns.

Formula: $C[i][j] = A[i][j] + B[i][j]$

2. Matrix Subtraction
Condition: Both matrices must have the same dimensions.

Formula: $C[i][j] = A[i][j] - B[i][j]$

3. Matrix Multiplication
Condition: Number of columns of the first matrix must be equal to the number of rows of the second matrix.

Formula: $C[i][j] = \Sigma (A[i][k] \times B[k][j])$ where $k = 0$ to $n-1$

4. Transpose of a Matrix
The transpose of a matrix is obtained by interchanging its rows and columns.

Formula: AT[i][j] = A[j][i]

These operations are implemented using two-dimensional arrays, nested loops, and conditional statements.

IMPLEMENTATION:

- **PLATFORM:**
    - 64-bit Open source Linux or its derivatives.
    - Open Source C Programming tool like gcc/Eclipse Editor.
- **PSEUDO CODE:** ***//To be Written by Students***

Write pseudo code for matrix operations as addition, subtraction, multiplication, and transpose.

- **TIME COMPLEXITY:** ***//To be Written by Students***

Find out the time complexity of all matrix operations

- **CONCLUSION:** ***//To be Written by Students***
- **Frequently Asked Questions (Descriptive)** ***//To be Written by Students***

1. Explain matrix addition, subtraction, multiplication, and transpose with suitable examples and state the conditions required for each operation.

2. Describe how two-dimensional arrays are used to represent matrices in C programming. Explain the process of accessing and manipulating matrix elements using nested loops.

3. Explain why compatibility between matrices is necessary before multiplication.

4. Discuss the applications of matrix operations in real-world domains such as computer graphics, image processing, machine learning, engineering, and scientific computing.

5. Explain the role of nested loops in implementing matrix operations in C. Illustrate how nested loops are used for addition, multiplication, and transpose of matrices.